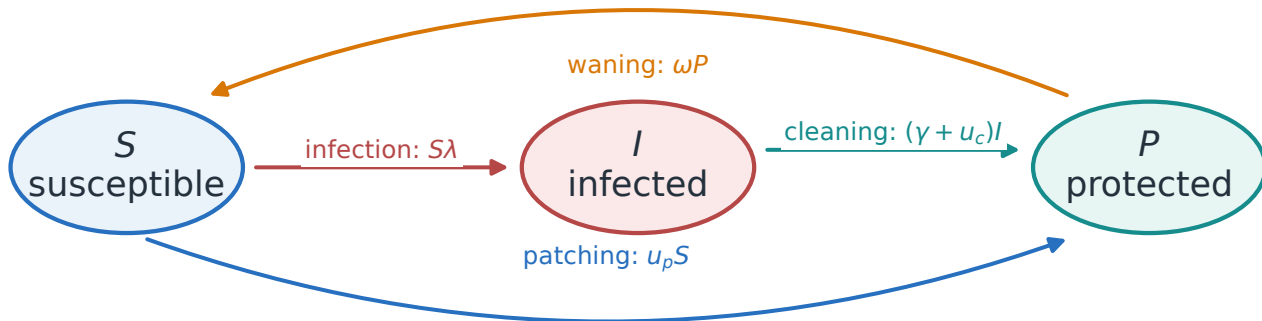


SIR and SIPS state transitions

Aggregate SIR uses $[S, I, R]$; node control uses $[S_i, I_i, P_i]$. Both conserve total mass.



Aggregate SIR: $[S, I, R]$
 $S \rightarrow I: \beta SI; I \rightarrow R: \gamma I$
optional control moves S or I to R ; $S + I + R = 1$

Node-indexed pressure
$$\lambda_i = \beta_i s_i \sum_j A_{ij} \iota_j I_j$$

node-specific $s_i, \iota_i, \gamma_i, \omega_i$
and node-specific control bounds

The invariant is a modeling check, not an extra loss term in the ODE solver.